



Proof by Induction · Mark Scheme

Core Pure 1 Chapter 8: Proof by Induction. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Summation formulae

1 $n = 1$: LHS = 1, RHS = $\frac{1}{2}(1)(2) = 1$, so true for $n = 1$ B1; assume true for $n = k$: $\sum_{r=1}^k r = \frac{1}{2}k(k+1)$ B1.

2 $\sum_{r=1}^{k+1} (2r-1) = k^2 + (2(k+1)-1)$ M1; $= k^2 + 2k + 1$ A1; $= (k+1)^2$ A1.

3 (a) $\frac{1}{2}(k^2 + k + 2) + (k+1) = \frac{1}{2}(k^2 + 3k + 4)$ M1; $= \frac{1}{2}((k+1)^2 + (k+1) + 2)$ A1. (b) The basis fails: for $n = 1$, LHS = 1 but RHS = 2, so the chain never starts B1.

4 $n = 1$: LHS = 2, RHS = $\frac{1}{3}(1)(2)(3) = 2$ B1; assume true for $n = k$ and add the $(k+1)$ th term: $\frac{1}{3}k(k+1)(k+2) + (k+1)(k+2)$ M1; $= \frac{1}{3}(k+1)(k+2)(k+3)$ A1; this is the formula with $n = k+1$; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

5(a) $n = 1$: LHS = 1, RHS = $\frac{1}{4}(1)(4) = 1$ B1; assume true for $n = k$: $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}k^2(k+1)^2 + (k+1)^3$ M1; $= \frac{1}{4}(k+1)^2(k^2 + 4k + 4)$ M1; $= \frac{1}{4}(k+1)^2(k+2)^2$ A1; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

5(b) $\sum_{r=1}^{2n} r^3 - \sum_{r=1}^n r^3 = \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{4}n^2(n+1)^2$ M1; $= \frac{1}{4}n^2[(4n+2)^2 - (n+1)^2]$ M1; $= \frac{1}{4}n^2(3n+1)(5n+3)$ (difference of two squares) A1.

Section B – Divisibility

6 $f(k+1) = 5 \times 5^k - 1$ M1; $= 5(5^k - 1) + 4 = 5f(k) + 4$ A1.

7 $f(1) = 4$, divisible by 4 B1; assume $f(k) = 4M$; then $f(k+1) = 20M + 4 = 4(5M + 1)$ M1; divisible by 4; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

8 $f(1) = 9 + 7 = 16 = 8 \times 2$ B1; assume $f(k) = 8M$; $f(k+1) - f(k) = 9 \times 3^{2k} - 3^{2k}$ M1; $= 8 \times 3^{2k}$, so $f(k+1) = 8M + 8 \times 3^{2k} = 8(M + 3^{2k})$ A1; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

9 $f(1) = 3$ B1; $f(k+1) = (k+1)^3 + 2(k+1) = k^3 + 3k^2 + 5k + 3$ M1; $= f(k) + 3(k^2 + k + 1)$, a sum of multiples of 3 A1; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

10(a) $f(k+1) = 81 \times 3^{4k} + 16 \times 2^{4k+2}$ M1; subtracting $16f(k) = 16 \times 3^{4k} + 16 \times 2^{4k+2}$ gives 65×3^{4k} A1.

10(b) $f(1) = 81 + 64 = 145 = 5 \times 29$ B1; assume $f(k) = 5M$; $f(k+1) = 16f(k) + 65 \times 3^{4k}$ M1; $= 5(16M + 13 \times 3^{4k})$ A1; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

10(c) $f(2) = 3^8 + 2^{10} = 7585$, and $7585 \div 29 = 261.55 \dots$, so not always divisible by 29 B1.

Section C – Matrix powers

11 $M^2 = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$, $M^3 = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$ M1; $M^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ A1.

12 $n = 1$: $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = M$ B1; assume true for $n = k$; $M^{k+1} = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ M1; $= \begin{pmatrix} 1 & 2k+2 \\ 0 & 1 \end{pmatrix}$, true for $n = k+1$; true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

13 $n = 1$: $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ B1; assume true for $n = k$; $\begin{pmatrix} 2^k & 0 \\ 2^k-1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ M1; $= \begin{pmatrix} 2^{k+1} & 0 \\ 2^{k+1}-2+1 & 1 \end{pmatrix} = \begin{pmatrix} 2^{k+1} & 0 \\ 2^{k+1}-1 & 1 \end{pmatrix}$ A1; conclusion sentence as in Question 12 A1.



$$\begin{aligned}
 \mathbf{14} \quad n=1: & \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} B1; \text{ assume true for } n=k; \begin{pmatrix} 1+2k & -4k \\ k & 1-2k \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} M1; \\
 & = \begin{pmatrix} 3+2k & -4-4k \\ 1+k & -1-2k \end{pmatrix} = \begin{pmatrix} 1+2(k+1) & -4(k+1) \\ k+1 & 1-2(k+1) \end{pmatrix} A1;
 \end{aligned}$$

true for $n=1$, and true for $n=k$ implies true for $n=k+1$, so true for all $n \in \mathbb{Z}^+$ A1.

$$\begin{aligned}
 \mathbf{15(a)} \quad n=1: & \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix} = \mathbf{M} B1; \text{ assume true for } n=k; \mathbf{M}^{k+1} = \begin{pmatrix} 1+4k & -8k \\ 2k & 1-4k \end{pmatrix} \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix} M1; \\
 & = \begin{pmatrix} 5+4k & -8-8k \\ 2+2k & -3-4k \end{pmatrix} A1; = \begin{pmatrix} 1+4(k+1) & -8(k+1) \\ 2(k+1) & 1-4(k+1) \end{pmatrix} A1; \text{ true for } n=1, \text{ and true for } n=k \text{ implies true for } n=k+1, \\
 & \text{so true for all } n \in \mathbb{Z}^+ A1.
 \end{aligned}$$

15(b) $(1+4n)(1-4n) + 16n^2 = 1 - 16n^2 + 16n^2 = 1$ B1; the transformation $\mathbf{M}^n$ preserves area (and orientation) B1.

$$\mathbf{15(c)} \quad \mathbf{M}^{-1} = \begin{pmatrix} -3 & 8 \\ -2 & 5 \end{pmatrix} B1; n=-1 \text{ gives } \begin{pmatrix} 1-4 & 8 \\ -2 & 1+4 \end{pmatrix}, \text{ which is the same } B1.$$

Roar-level challenge

$$\begin{aligned}
 \mathbf{16} \quad n=1: & \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}, \text{ true } B1; \text{ assume true for } n=k; \mathbf{M}^{k+1} = \mathbf{M}^k \mathbf{M} M1; \text{ top row: } 3(2^{k+1}-1) + (2-2^{k+1}) = 2^{k+2}-1 \\
 & \text{and } -2(2^{k+1}-1) = 2-2^{k+2} A1; \text{ bottom row: } 3(2^k-1) + 2-2^k = 2^{k+1}-1 \text{ and } -2(2^k-1) = 2-2^{k+1} A1; \text{ true for } n=1, \\
 & \text{and true for } n=k \text{ implies true for } n=k+1, \text{ so true for all } n \in \mathbb{Z}^+ A1.
 \end{aligned}$$